



Published in final edited form as:

J Bone Miner Res. 2021 July ; 36(7): 1387–1402. doi:10.1002/jbmr.4291.

Premature growth plate closure caused by a hedgehog cancer drug is preventable by co-administration of a retinoid antagonist in mice

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Abstract

The growth plates are key engines of skeletal development and growth and contain a top reserve zone followed by maturation zones of proliferating, pre-hypertrophic and hypertrophic/mineralizing chondrocytes. Trauma or drug treatment of certain disorders can derange the growth plates and cause accelerated maturation and premature closure, one example being anti-hedgehog drugs such as LDE225 (Sonidegib) used against pediatric brain malignancies. Here we tested whether such acceleration and closure in LDE225-treated mice could be prevented by co-administration of a selective retinoid antagonist, based on previous studies showing that retinoid antagonists can slow down chondrocyte maturation rates. Treatment of juvenile mice with an experimental dose of LDE225 for 2 days (100 mg/kg by gavage) initially caused a significant shortening of long bone growth plates, with concomitant decreases in: chondrocyte proliferation; expression of *Indian hedgehog*, *Sox9* and other key genes; and surprisingly, the number of reserve progenitors. Growth plate involution followed with time leading to impaired long bone lengthening. Mechanistically, LDE225 treatment markedly decreased the expression of retinoid catabolic enzyme *Cyp26b1* within growth plate whereas it increased and broadened the expression of retinoid synthesizing enzyme *Raldh3*, thus subverting normal homeostatic retinoid circuitries and in turn accelerating maturation and closure. All such severe skeletal and molecular changes were prevented when LDE-treated mice were co-administered the selective retinoid antagonist CD2665 (1.5 mg/kg/daily), a drug targeting retinoid acid receptor γ most abundantly

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Author contributions: **Eiki Koyama**: Conceptualization; data curation; formal analysis; methodology; visualization; validation; writing-review and editing. **Christina Mundy**: Investigation; data curation; formal analysis; methodology; visualization; writing-review and editing. **Cheri Saunders**: Investigation; validation. **Juliet Chung**: Investigation; validation. **Sarah E. Catheline**: Investigation; validation; data curation; formal analysis; visualization; writing-review and editing. **Danielle Rux**: Investigation; data curation; formal analysis; validation. **Masahiro Iwamoto**: Conceptualization; formal analysis; funding acquisition. **Maurizio Pacifici**: Conceptualization; data curation; formal analysis; project administration; funding acquisition; writing original draft; writing-review and editing.

Disclosures

The authors state that they have no conflict of interest with the study.

expressed in growth plate. When given alone, CD2665 elicited the expected maturation delay and growth plate expansion. In vitro data showed that LDE225 acted directly to dampen chondrogenic phenotypic expression, a response fully reversed by CD2665 co-treatment. In sum, our proof-of-principle data indicate that drug-induced premature growth plate closures can be prevented or delayed by targeting a separate phenotypic regulatory mechanism in chondrocytes. The translation applicability of the findings remains to be studied.

Keywords

Premature growth plate closure; chondrocyte maturation; hedgehog signaling drugs; retinoid antagonists and reverse agonists; reserve cell progenitors

Introduction

The growth plate is the key engine of skeletal development, growth and morphogenesis.^(1,2) Skeletal elements such as femur or tibia in young animals and individuals contain a growth plate at each of their proximal and distal ends, while elements such as ribs contain a single growth plate at the chondro-osseous junction. Regardless of location, growth plates display a stereotypic structure composed of multiple chondrocyte columns each of which is distinguishable in a top reserve cell zone followed by proliferative, pre-hypertrophic, hypertrophic and mineralizing chondrocyte zones.^(1,2) The reserve zone contains self-renewing progenitor/stem cells, and recent genetic cell lineage tracing studies from our group revealed for the first time that a single progenitor is responsible for generating all the chondrocytes present in a column.^(3,4) This finding has been confirmed in subsequent studies indicating also that reserve zone progenitors are demarcated by expression of parathyroid hormone-related protein (PTHrP).^(5,6) Chondrocyte behavior and function in the remaining zones are regulated by complex mechanisms and interplays involving: signaling proteins such as hedgehog, fibroblast growth factor (FGF) and bone morphogenetic protein (BMP) family members; transcription factors including Runx2 and Mef2c; and systemic hormones and other effectors.^(2,7-9) One particularly important mechanism consists of a regulatory feedback loop between PTHrP expressed in the top zone and Indian hedgehog (Ihh) expressed in the pre-hypertrophic zone.⁽¹⁰⁻¹³⁾ This relay system is critical for establishing and maintaining the rates at which chondrocytes proliferate and advance through the subsequent steps of maturation and hypertrophy, thus setting the overall pace of growth plate activity and skeletal element elongation.⁽¹⁾

Trauma or drug treatment of certain pathologies can have adverse side effects on overall skeletal growth, can derange growth plate's regulatory mechanisms and may lead to premature growth plate closure.^(14,15) For example, children affected by epilepsy are commonly treated with drugs such as valproic acid, oxcarbazepine or topiramate that can affect the skeleton.^(16,17) One study reported that some children receiving valproic acid for at least one year exhibited impairment of skeletal growth compared to a control group.⁽¹⁸⁾ The authors also tested rat growth plate chondrocytes in vitro and found that treatment with valproic acid caused a significant decrease in cell proliferation, leading them to conclude that the skeletal growth retardation in the valproic acid-treated epileptic children

was due to a derangement of growth plate chondrocyte proliferation. Pediatric patients suffering from medulloblastoma provide another example of drug-induced side effects on skeletal growth. About 25% of these cancers are dependent on local hedgehog signaling and action for cell growth and survival,⁽¹⁹⁾ and treatment with drug antagonists blocking the hedgehog signaling receptor Smoothed such as GDC0449 (vismodegit) and LDE225 (sonidegib) has proven very effective in reducing tumor growth and enhancing survival of medulloblastoma patients.^(20–22) However, studies reported that a few of the treated pediatric patients displayed skeletal growth retardation and premature growth plate closure, most likely caused by interruption of the Ihh-PTHrP relay system and consequent acceleration of chondrocyte maturation and hypertrophy.^(23,24) The patients exhibited also bending of skeletal elements such as the radius as well as bony protrusions, possibly reflecting aberration of normal growth plate roles in alignment, orientation and three-dimensional sculpting of skeletal elements.⁽²⁵⁾ It should be underlined that once a growth plate is fully closed, there are no means at present to reverse the process and reestablish its function. Thus, there is a need to identify possible therapeutics that could preserve the growth plate and avoid its premature closure during drug treatment of various pathologies.

These and other considerations led us to consider the retinoid signaling pathway, an additional and important growth plate regulatory mechanism in which we have a long-standing interest. This pathway involves natural ligands such as all-*trans*-retinoic acid (RA) that cells produce in response to physiologic needs and utilizing circulating inert precursors such as retinol released from liver storages.⁽²⁶⁾ Once produced by cytoplasmic enzymes, RA and other active retinoids such as 9-*cis*-RA are transported to the nucleus via carrier proteins where they interact with retinoic acid receptors (RARs) bound to RA-responsive elements (RAREs) and associated with RXRs, resulting in transcription of target genes.^(27,28) Notably and importantly, in the absence of active ligands, the RARE-bound RAR/RXR complexes operate as transcriptional repressors, and this function is essential for a variety of physiologic processes as well.^(27,29) In previous studies, we and others showed that the top zones of growth plate are largely devoid of endogenous active retinoids and rely on unliganded RAR repression function, while the lower zones do contain retinoids, likely favoring terminal chondrocyte maturation and hypertrophy and transition to endochondral bone.^(30–32) Of particular relevance to the present study was also our finding that when RAR repressor function was pharmacologically enhanced by treatment with RAR antagonists (acting as inverse agonists),⁽³³⁾ it further slowed down chondrocyte maturation and hypertrophy and transition to bone.⁽³⁴⁾ These and other considerations raised the possibility that co-administration of a retinoid antagonist could be able to slow down or even prevent the premature chondrocyte hypertrophy and growth plate closure caused by cancer drugs such as LDE225, thus preserving growth plate function and maintaining skeletal elongation. The proof-of-principle data reported here provide strong evidence that this is in fact the case, at least in mice.

Materials and Methods

Ethics statement regarding mouse studies

All studies involving wild type and transgenic mice were reviewed by our institutional IACUC at The Children's Hospital of Philadelphia. All animals were handled, treated and cared for according to the approved protocols, and all procedures fully complied with the ARRIVE guidelines. CD1 and *Pthrp^{mCherry/+}* mice (JAX032872) were obtained from the Jackson Laboratory (Bar Harbor, ME, USA).

Drug administration

Stock solutions of the hedgehog antagonist LDE225 (ChemieTek, Indianapolis, IN, USA) were prepared in DMSO at 100 mg/ml final concentration. Stock solutions of the selective retinoic acid receptor antagonist CD2665 (Tocris Bioscience, Minneapolis, MN, USA) were also prepared in DMSO at 100 mM final concentration. CD2665 has a K_i value of 110 nM for RAR γ and 306 nM for RAR β .⁽³⁵⁾ Stocks were subdivided into small aliquots and stored at -30°C until use. On days of treatment, a sufficient number of aliquots were thawed and diluted with a mixture of DMSO/corn oil mixture (3:7) to working concentration. Mice were administered drugs by daily gavage at 100 mg/kg LDE225 and/or 1.6 mg/kg CD2665 in a total of 100 to 200 μl /gavage. Control mice were given the same volume of DMSO/corn oil mixture.

Sufficient numbers of male and female mice per experiment were placed in separate cages and randomized into control and experimental groups to minimize bias. Mice were sub-divided into 4 subgroups and treated with LDE225, CD2665, LDE 225/CD2665 combination or vehicle. Each subgroup contained a minimum of 5 mice per experiment. Drug treatment started at 3 weeks of postnatal age (P21). Mice receiving LDE225 only were treated on P22 and P23. Mice receiving the combination drug regimen were pre-treated with CD2665 on P20 and P21, were given both LDE225 and CD2665 on P22 and P23 and were then given a daily dose of CD2665 until end of experiment. Mice receiving CD2265 alone were given the drug daily throughout the period used as were companion controls receiving daily vehicle administration. Before gavage, mice were anesthetized by inhalation of 1.5% isoflurane in 98.5% oxygen to minimize discomfort as prescribed by IACUC. Oral gavage probes were 22 gauge and 25 mm long (Instech Laboratories, Plymouth Meeting, PA, USA). Following drug administration, animals were regularly monitored for signs of overt toxicity including lethargy and dystonia, and Institutional LAS personnel provided additional daily animal monitoring and care.

X-ray, histological, and histochemical analyses

To analyze long bone length and growth plate status, mouse X-ray radiographs were taken with 20 s exposures at 25kV (Faxitron UltraFocus 100, Tucson, AZ, USA). Resulting data were used to calculate skeletal length averages and statistical significance as indicated below. After imaging, limbs were harvested, fixed overnight in 4 % paraformaldehyde (PFA) and decalcified in 10% EDTA/1% PFA for 2 weeks. Samples were dehydrated and embedded in paraffin, and continuous 5 μm sections were stained with hematoxylin or Safranin-O/fast green for histological and histomorphometric evaluations. To calculate

growth plate length and chondrocyte zone heights, images of one in every 6 consecutive sections encompassing the entire mid-frontal region of tibial growth plates were analyzed using ImageJ software. To define growth plate chondrocyte phenotype at indicated mouse ages, continuous sections were processed for in situ hybridization as detailed below.

To analyze slow-cycling cells, the thymidine analogue 5-ethynyl-2'-deoxyuridine (EdU) (A10044, ThermoFisher Scientific, Branchburg, NJ, USA) was administered 5 times by serial daily intraperitoneal injections (at 100 mg/kg) from P5 to P9. A minimum of 3 mice per group were used in each experiment. Mice were sacrificed at P27, and limbs were fixed and decalcified as above and processed for EdU detection. Click-iT EdU AlexaFluor imaging kit reaction (Invitrogen, Carlsbad, CA, USA) was used according to manufacturer's protocol, and sections were mounted with Fluor-Gel II Mounting Medium. Images of one in every 6 sections encompassing the entire mid-frontal region of tibial growth plate were analyzed and quantified by ImageJ.

For the detection of *Pthrp* reporter-positive cells, mouse limbs of each group were fixed on P27 and decalcified for 1 week as above. Specimens were then washed with 1x PBS, equilibrated in 30% sucrose in PBS overnight at 4°C, and mounted in OCT. Frozen 12 µm-thick frontal-oriented sections were prepared by cryostat, using Kawamoto's cryofilm method.⁽³⁶⁾ Six sections collected from mid-frontal region of each tibial growth plate were analyzed per sample, and *Pthrp*-expressing were imaged and counted using a Nikon fluorescence scope.

In situ hybridization

Paraffin sections were processed for in situ hybridization using radioisotope-labeled riboprobes or RNAscope technology as indicated. Briefly, after paraffin removal and rehydration, serial tissue sections were pretreated with 10 µg/ml proteinase K for 10 min at room temperature, post-fixed in 4% paraformaldehyde, washed with 1x PBS containing 2 mg/ml glycine as quencher, and treated with 0.25% acetic anhydride in triethanolamine buffer.⁽³⁷⁾ Sections were hybridized with antisense or sense ³⁵S-labeled riboprobes (approximately 1×10⁶ DPM/section) at 50°C for 16 h. After hybridization, sections were washed with 2xSSC containing 50% formamide at 50°C, treated with 20 µg/ml RNaseA for 30 minutes at 37°C, washed with 0.1XSS, coated with Kodak NTB-3 emulsion diluted 1:1 with water, and exposed for 3 days for detection of *Collagen II* (*Col II*) expression, 7 days for *osteopontin* expression, and 14 days for *Sox9* or *histone H4C* (*H4C*) expression. Sections were developed with Kodak D-19 at 18°C for 3 min and counter-stained with hematoxylin. Dark- and bright-field images were captured using a digital camera, and dark-field images were pseudo-colored using Adobe Photoshop software. cDNA clones used as templates for riboprobe production included: *Sox9* (nucleotides (nt.) 116–856; NM_011448), *Col II* (nt. 1095–1344; X57982), *Col X* (nt.1302–1816; NM009925), *osteopontin* (*Spp1*; nt.115–593; AF515708), and *H4C* (nt. 549–799; AY158963).

RNAscope in situ hybridization was carried out using RNAscope2.5 HD Detection reagent-RED (Advanced Cell Diagnostics, Newark, CA, USA) to visualize the spatio-temporal expression of *Col X* (cat. no. 467961), *Ihh* (cat. no. 413091), *cytochrome P450 family 26-b1* (*Cyp26b1*) (cat. no. 454241) and *retinaldehyde dehydrogenase 3* (*Raldh3*) (cat. no. 501201)

in tibial growth plates. Briefly, growth plate sections were pretreated with a custom reagent and hybridized with each probe for 2 hrs at 40°C in a custom oven. Signal was amplified with pre-amplifier and multiple amplifier as per manufacturer's protocols, and final signal was detected and visualized by reaction with Fast Red substrate for about 10 min-incubation at room temperature. Companion sections were hybridized with positive (Cat No. 313911) or negative control probes (Cat. No. 310043) to assure signaling specificity. Sections were counterstained with hematoxylin, dried and sealed with Permount.

Preparation, treatment and analysis of micromass cultures

Micromass culture were prepared from E11.5 CD-1 mouse embryos limb buds as described.⁽³⁸⁾ Briefly, limb buds was dissociated in single cell suspensions with 0.5% trypsin-EDTA incubation at 37°C for 20 min. Dissociated cells were re-suspended at a concentration of 10×10^6 cell/ml in DMEM containing 10% fetal bovine serum and antibiotics. Micromass culture was initiated by spotting 15 μ l of the cell suspensions (1.5×10^5 cells/spot) on the surface of 24-well tissue culture plates. After a 2 h incubation at 37°C in a humidified CO₂ incubator to allow for cell attachment, cultures were given 500 μ l of medium. After 24 h, medium supplemented with LDE225 (5 μ M), CD2665 (0.3 nM) or a combination of the two was added to the cultures. Fresh reagents were given with medium change every other day. Equivalent amount of vehicle (DMSO) was given to control cultures. Cultures were stained with alcian blue to detect cartilage nodule formation. Images were taken by a Nikon SMZ-U microscope equipment with a SPOT insight camera (Diagnostic Instruments Inc., Sterling Heights, MI, USA) and acquired with SPOT 4.0 software. Micromass analysis was performed using ImageJ. Images were made binary under an RGB threshold and Particle Analysis was utilized to measure alcian blue positive area.

Total RNA was isolated from control and drug treated micromass cultures on day 3 and 6 using TRIzol reagent according to the manufacturer's protocol and quantified by Nanodrop. One microgram total RNA was reverse-transcribed using the Verso cDNA kit (Life Technologies: Thermo Scientific, Newark, CA). Quantitative real-time PCR was carried out using SYBR Green PCR master mix in an Applied Biosystems 7500 according to manufacturer's protocol. *Gapdh* was used as endogenous housekeeping control, and relative expression level of *Sox9*, *aggrecan (Acan)* and *Col II* was calculated using the $\Delta\Delta C_t$ method. Multiple samples were analyzed in each experiment, and experiments were repeated independently a minimum of 3 times. Sequences of forward and reverse PCR primers used are presented in Supplemental Table S1.

Statistical procedures

Statistical significance of data from experimental groups was evaluated by one-way ANOVA followed by Tukey's multiple comparison test. Data were plotted using GraphPad Prism 7 (GraphPad Software, San Diego, CA, USA), and data points, averages and 95% confidence intervals are included in scatter plots presented below. A statistical difference of $p < 0.05$ was considered significant, and p values shown in the figures denote degrees of statistical significance. Representative images were used to illustrate observations, and number of animals used in each experiment is specified in the respective figure legend. Cell culture

experiments were repeated independently at least three times, and data from a representative experiment are presented.

RESULTS

Acute growth plate closure in juvenile mice is preventable by drug co-treatment

In order to establish an effective and rapid experimental model of premature growth plate closure for our proof-of-principle studies, we used the hedgehog antagonist LDE225 (sonidegib). The drug is in clinical use for conditions such as basal cell carcinoma and is in general safe at currently prescribed doses in adults.⁽³⁹⁾ As pointed out above, the drug has also been found to cause skeletal growth retardation in a small number of pediatric medulloblastoma patients^(23,24) as well as rapid growth plate closure in juvenile mice when given at excess experimental research doses.⁽⁴⁰⁾ Thus, 3 week-old male and female CD-1 mice were administered a daily oral experimental dose of LDE225 at 100 mg/kg in corn oil for 2 days total (postnatal days 22 and 23; P22 and P23) (Fig. 1A). Companion mice received the same regimen of vehicle (corn oil). After drug administration, the mice were allowed to resume activities and did not exhibit obvious changes in behavior or fur coat appearance over the duration of the study (see Fig. 2B below). However, histological examination of long bones harvested at P24, P27 and P31 (1, 4 and 8 days from the last LDE225 administration, respectively) revealed the occurrence of deleterious effects on growth plate architecture (Fig. 1). In control mice receiving vehicle, the growth plates in tibia or femur displayed their stereotypic organization in which the major resting (*re*), proliferative (*pl*), pre-hypertrophic (*ph*) and hypertrophic/mineralizing (*hy*) zones were easily recognizable (Fig. 1B–E). Instead, the tibial growth plate in LDE225-treated mice already displayed a reduction in height by P24 and P27 (Fig. 1F–H) and was essentially closed by P31 (Fig. 1I). Histomorphometric quantifications showed that the overall growth plate height as well as the height of pre-hypertrophic and hypertrophic zones height (Fig. 1C vs 1G–H, vertical yellow bar) were progressively reduced by drug treatment until full closure by P31 (Fig. 1I and R–S).

To determine whether such rapid LDE225-induced growth plate involution could be prevented or at least ameliorated by drug co-administration, we tested the efficacy of the synthetic RAR antagonist CD2665 that preferentially inhibits RAR γ ⁽³⁵⁾ increasing RAR repressor function.⁽³³⁾ We chose a selective antagonist rather than a global RAR pan-antagonist because RAR γ is the major RAR family member expressed in growth plate^(32,34) and because a compound antagonizing all the RARs may have unwanted side effects. As above, mice were given a dose of LDE225 at P22 and P23, but were also pre-administered the RAR γ antagonist at 1.5 mg/kg at P20 and P21 and thereafter daily until P30 (Fig. 1A). The antagonist dose was chosen based on original mouse effectiveness studies.⁽⁴¹⁾ Remarkably, growth plate closure had largely been prevented in LDE225/RAR γ antagonist co-treated mice compared to those receiving LDE225 only, evident both visually (Fig. 1J–M) and by histomorphometry (Fig. 1R–S). As expected, companion mice receiving RAR γ antagonist alone displayed an appreciable delay in chondrocyte maturation and ossification indicated by a slight increase in overall growth plate height compared to control at P31

(Fig. 1O vs 1E) and an expansion of *Col II*-expressing upper zones and *Col X*-expressing hypertrophic zone (Supplemental Fig. 1J–L vs A–C).

To determine whether the changes in growth plate had repercussions on overall skeletal growth and elongation, groups of companion male and female P20 mice receiving vehicle, LDE225, RAR γ antagonist or LDE225/RAR γ antagonist combination exactly as above (Fig. 2A) were monitored for body length and body weight over time. The general appearance, behavior and fur coat were not overtly different in treated versus control mice of either gender during the short time course of these experiments (Fig. 2B–C). However, there was a significant decrease in both body length and weight in LDE225-treated versus control mice, amounting to about 10–15% decrease by P31 (Fig. 2D–G). Notably, such deficits were largely, albeit not completely, rescued by RAR γ antagonist co-administration (Fig. 2D–G). Whole body X-ray imaging confirmed that tibia length was appreciably reduced by LDE225 treatment, and this trait also was partially rescued by RAR γ antagonist co-treatment (Fig. 2H–J). The X-ray images showed that the tibia growth plate was open and clearly visible in control as well as LDE225/RAR γ antagonist co-treated mice, while it was reduced in LDE225-treated mice (Fig. 2H, arrowheads in green box insets). All the above skeletal parameters were essentially unaffected in companion mice receiving RAR γ antagonist alone (Fig. 2). Given the paucity of responses, this treatment regimen was not included in subsequent experiments.

Growth plate gene expression patterns are rectified by retinoid antagonist co-administration

The beneficial effects of RAR γ antagonist co-administration on growth plate and skeletal growth above suggested that the drug could have also normalized the growth plate at the phenotypic and molecular levels. To assess this prediction, serial longitudinal sections of tibias harvested at P31 from control, LDE225-treated and LDE225/RAR γ antagonist co-treated mice (Fig. 3A) were processed for *in situ* hybridization to analyze the expression of key marker genes for different growth plate zones.⁽²⁾ Control growth plates from vehicle-receiving mice exhibited typical and strong expression of: *Sox9* and collagen II (*Col II*) in proliferative and pre-hypertrophic zones; collagen X (*Col X*) in the hypertrophic zone; and *osteopontin* in mineralizing zone and primary spongiosa (Fig. 3B–E). In comparison, expression of *Sox9*, *Col II* and *Col X* was virtually undetectable in tibia sections from P31 LDE225-treated mice (Fig. 3F–H), while *osteopontin* expression was widespread and scattered across the bony tissues now occupying the epiphyseal-metaphyseal region and replacing the growth plate (Fig. 3I). As clearly shown in Fig. 3J–M, the gene expression patterns of *Sox9*, *Col II*, *Col X* and *osteopontin* had been largely preserved in LDE225/RAR γ antagonist co-treated mice.

Analysis of chondrocyte proliferation and reserve zone cells

Chondrocyte proliferation is an important component of overall growth plate contribution to skeletal growth and elongation, and its rates are regulated by the Ihh/PTHrP loop.⁽¹⁾ The subpar skeletal elongation observed after LDE225 treatment and the recovery seen after LDE225/RAR γ antagonist co-treatment suggested that chondrocyte proliferation might have been greatly inhibited in the former condition, but maintained after drug co-treatment. To

test these theses, mice were subjected to regimens of vehicle or single/compound drug administration starting at P20 exactly as above (Fig. 4A) and were then sacrificed at P27 when the growth plates were still appreciable in all conditions (see Fig. 1). Serial sections of tibia were then processed for expression analysis of the cell mitotic marker histone 4C (*H4C*)⁽⁴²⁾ in relation to *Ihh* expression. In controls, *H4C* expression was strong and characterized chondrocytes in the proliferative zone and also cells within the primary spongiosa (Fig. 4B), while *Ihh* was readily detectable in the pre-hypertrophic zone (Fig. 4C). Proliferation patterns were preserved both quantitatively and topographically (growth plate and primary spongiosa) in LDE225/RAR γ antagonist co-treated mice (Fig. 4F–I), but had been wholly compromised in LDE225-treated mice (Fig. 4D–E and 4H–I). In the latter, the near absence of proliferative cells in both growth plate and primary spongiosa indicates that hedgehog action coordinates proliferation at both sites.

As pointed out above, the reserve zone of growth plate contains slow-cycling progenitor stem cells that are engaged in cell renewal and also replenish the growth plate over time by providing newly committed chondrocytes.^(3,4) Reserve zone cells are usually identified by a pulse-chase strategy using the thymidine analogue 5-ethynyl-20-deoxyuridine (EdU) followed by a protracted chase period.⁽⁴³⁾ Accordingly, mice received a daily systemic administration of EdU by intraperitoneal injection from P5 to P10 (100 mg/kg EdU/injection) (Fig. 5A). Mice were then chased until P20 by which time EdU incorporated in proliferating chondrocytes would be expected to largely dissipate through ongoing mitotic activity, while EdU incorporated into slow cycling reserve progenitors would still be detectable. The P20 mice were then subdivided into 3 groups, received vehicle, LDE225 or LDE225/RAR γ antagonist combination starting at P21 exactly as above, and were then sacrificed at P27. Immunohistochemistry did show that EdU-labeled cells were present and largely confined to the reserve zone of control tibial growth plate (Fig. 5B–C, bracket). Their number was greatly diminished in the growth plate in LDE225-treated mice (Fig. 5E–F), but had been preserved in LDE225/RAR γ antagonist co-treated mice (Fig. 5H–I, bracket). On average, the top zone contained an average of 15–17 labeled cells/5 randomly chosen microscopic fields in controls, and this number had been reduced to about 2–3 cells/5 fields in LDE225-treated mice and maintained at about 10–11 cells/5 fields in co-treated mice (Fig. 5K). To verify these findings, we monitored skeletal progenitors in the reserve zone of growth plate by virtue of their *Pthrp* expression.⁽⁶⁾ Thus, *Pthrp*^{mCherry/+} knock-in mice (*Pthrp* reporter mice) were subdivided into 3 groups that received vehicle, LDE225 or LDE225/RAR γ antagonist combination exactly as above. *Pthrp* reporter-positive cells were largely confined to the reserve zone of control tibial growth plate (Fig. 5D). Their number was greatly diminished in the growth plate in LDE225-treated mice (Fig. 5G), but was preserved in LDE225/RAR γ antagonist co-treated mice (Fig. 5J). There were an average of 13–15 *Pthrp* reporter expressing cells/5 randomly chosen microscopic fields in controls, and this number had been reduced to less than 2–3 cells/5 fields in LDE225-treated mice but maintained at about 8–11 cells/5 fields in co-treated mice (Fig. 5L).

Expression topography of the retinoid metabolic enzymes during drug treatment

Physiologic levels of active retinoids in cells are tightly regulated by the opposing activities of intracellular retinoid-synthesizing retinaldehyde dehydrogenase (RALDH)

family members and retinoid-catabolizing cytochrome P450 family 26 (CYP26) members.⁽⁴⁴⁾ In previous studies, we and other groups found that growth plate upper zones in postnatal animals are largely devoid of endogenous metabolically active retinoids such as all-*trans*-RA and 9-*cis*-RA, whereas the hypertrophic zone does contain active retinoids that promote transition to endochondral bone.^(30–32) Thus, we asked whether the premature growth plate closure induced by LDE225 might involve -and be possibly due to- changes in gene expression in *Cyp26* and *Raldh* family members, thus altering retinoid homeostasis and deranging chondrocyte behavior. To test this thesis, P20 mice were subjected to vehicle, LDE225 treatment or LDE225/RAR γ antagonist co-treatment as above and their hindlimbs were harvested at P24 and P27 and processed for *in situ* hybridization, using serial longitudinal tibia sections (Fig. 6A). We initially analyzed all family members *Raldh1*, *Raldh2* and *Raldh3*⁽⁴⁴⁾ and *Cyp26a1*, *Cyp26b1* and *Cyp26c1*⁽⁴⁵⁾, but found that the postnatal mouse growth plates mainly expressed *Raldh3* and *Cyp26b1*. *Cyp26b1* expression characterized the upper zones of control P24 and P27 growth plates (Fig. 6B, N), whereas *Raldh3* expression was restricted to the hypertrophic zone (Fig. 6C, O) overlapping *collagen X* expression (Fig. 6D–E and P–Q). In P24 growth plates from LDE225-treated mice, *Cyp26b1* expression was sharply decreased (Fig. 6F), while *Raldh3* expression was increased and much broader, overlapping the thicker *Col X*-expressing hypertrophic zone (Fig. 6F, H–I, vertical brackets). By P27, expression of *Cyp26b1* and *Raldh3* was barely appreciable given the dramatic involution of growth plate by then (Fig. 6R, S), with just a few *Col X*-expressing chondrocytes remaining visible (Fig. 6T–U). It was very interesting to observe that nearly normal patterns of *Cyp26b1*, *Raldh3* and *Col X* gene expression had been preserved in the growth plates of LDE225/RAR γ antagonist co-treated mice at either time point (Fig. 6J–M and V–Y).

Chondrogenic cells respond directly to drug treatment and co-treatment

The ability of the RAR γ antagonist to prevent growth plate closure caused by LDE225 is likely due to direct action of the drugs on chondrocytes, though it could also entail indirect systemic mechanisms. Previously we showed that another anti-hedgehog signaling drug acted directly on chondrogenic cells in vitro and suppressed their differentiation,⁽³⁸⁾ whereas treatment with RAR antagonists boosted it.⁽⁴⁶⁾ Thus, we asked whether LDE225 and RAR γ antagonist exerted similar anti- and pro-chondrogenic action, respectively and whether co-treatment counteracted LDE225 effects. Accordingly, primary limb bud mesenchymal cells isolated from E11.5 mouse embryos were reared in high density micromass cultures in standard medium or medium containing LDE225 or drug combination. Fresh drugs were added daily along with a medium change, and effects were analyzed after 4 to 6 days. In control vehicle-receiving cultures, many cartilage nodules had formed by day 6 and were appreciable after staining with alcian blue (Fig. 7A). Treatment with LDE225 or RAR γ antagonist did result in sharp decreases and increases in nodule formation and alcian blue levels, respectively (Fig. 7B, E). Such opposing responses were paralleled by declines or increases in expression of chondrogenic marker genes *Sox9*, *aggrecan* and *Col2a1* as revealed by qPCR (Fig. 7F–H). Indeed, co-treatment with RAR γ antagonist counteracted the LDE225 effects, and the co-treated cells displayed normalized levels of cartilage nodules, alcian blue staining levels and expression of chondrogenic markers (Fig. 7D–H). To gain insights into mechanisms of phenotypic expression, we extended the qPCR analyses to

regulators of chondrocyte maturation and terminal differentiation such as *Runx2* and *Mef2c* (47,48) and to transducers of hedgehog signaling.⁽⁴⁹⁾ LDE225 treatment left the expression of *Runx2* and *Mef2c* as well as *ColX* largely unaffected (Fig. 7I–J) while inhibiting *Ptc1* and *Gli1* expression (Fig. 7M–N), indicating minor roles of hedgehog signaling on expression of terminal maturation markers. In comparison, RAR γ antagonist treatment significantly increased the expression of *Runx2*, *ColX*, *Ptc1* and *Gli1* while concurrently decreasing *Mef2c* expression (Fig. 7I–K and M–N), suggesting that the overall chondrocyte maturation program had been delayed by the RAR γ antagonist as it was in vivo (see Fig. 1). Co-treatment reversed many but not all of LDE225 effects, signifying that normal balancing interactions between the two pathways had not been fully restored and that modulation of retinoic signaling may be more penetrant phenotypically. Notably, *Ihh* expression was significantly and consistently increased by both single and combined treatments (Fig. 7L). Given IHH's central role in growth plate maturation and functioning,⁽¹⁾ such response was indicative of an effort by the cells to maintain signaling and phenotypic normalcy during drug treatment.

Discussion

The data we present here provide clear and novel evidence that systemic co-administration of a retinoid antagonist can prevent the deleterious effects on growth plates caused by experimental doses of LDE225. The co-treatment allows the growth plates to maintain their morphological organization, phenotypic expression and contribution to skeletal elongation, at least within the time frames studied. As pointed out above, LDE225 and other similar drugs are eliciting measures of therapeutic success against malignancies such as hedgehog-dependent pediatric medulloblastomas that are extremely severe and usually fatal pathologies.^(20–22) Given the fact that the hedgehog signaling pathway regulates a very large number of developmental and physiologic processes including growth plate functioning,⁽⁵⁰⁾ it is not at all surprising that anti-hedgehog drugs could have unwanted side effects in a few of the treated patients. While data from mouse studies may or may not necessarily apply to patients, our present proof-of-principle data are encouraging and provide support for a possible drug combination strategy⁽⁵¹⁾ to prevent or at least reduce side effects on growth plate caused by hedgehog cancer drugs.

One of the most dramatic responses observed in our study is the rapidity with which the long bone growth plates underwent precocious closure after the start of LDE225 treatment at the doses used. Original cell kinetics studies revealed that the average half-life of chondrocytes in the growth plate is 4 to 6 days,^(52,53) corresponding to the time period required for chondrocytes in the proliferative zone to transit through the pre-hypertrophic, hypertrophic and mineralizing zones before undergoing involution or participating in primary spongiosa formation.⁽⁵⁴⁾ Differences in chondrocyte kinetics were noted in the proximal vs distal growth plates of long bones,⁽⁵⁵⁾ but they are not dramatic and still point to the overall ephemeral nature of the cells. These and other studies provide an explanation as to why the overall height of growth plates was already reduced after a few days of LDE225 treatment, delineating how dependent the growth plate must normally be on continuous hedgehog signaling. Given the known essential roles of the *Ihh*-PTHrP signaling loop in determining the rates of proliferation and maturation of growth plate chondrocytes, it is likely then that

LDE225 interference with such loop would have caused a rapid depletion of proliferating chondrocytes and rapid involution of the growth plate. But what could explain the observed rapid depletion of EdU-labeled slow cycling reserve progenitors and what could account for the ability of the RAR γ antagonist to rescue them?

The presence of reserve progenitors in the growth plate has been known for quite some time^(56,57), but only recently have studies begun to define their phenotypic characteristics and regulatory mechanisms, including *Pthrp* expression itself and Tsc1 signaling.^(5,6) Tellingly, one of these studies showed that conditional ablation of *Pthrp*-expressing resting cells by a diphtheria toxin fragment A (DTA) strategy altered the growth plate architecture and caused premature chondrocyte hypertrophy.⁽⁶⁾ Based on our data after LDE225 treatment, we can now add hedgehog signaling as another critical pathway to maintain slow cycling cells in the resting zone. Thus, the rapid involution of growth plate during LDE225 administration was likely caused not only by a decrease in mitotic activity in proliferative zone, but also a failure to maintain resting cells and replenish the growth plate. Given that the RAR γ antagonist was able to fully counter the deleterious effects of LDE225 and that the top zones of growth plate are normally devoid of endogenous retinoids,⁽³¹⁾ it appears that unliganded RAR repression function also is very important if not critical for behavior and phenotype of both resting cells and proliferative chondrocytes and can compensate for LDE225-induced hedgehog signaling deficiency.

These interpretations are in line with additional data here and previous studies showing that endogenous retinoids are kept under strict spatiotemporal control to sustain normal skeletal development and growth plate function.^(30,31,58) In agreement with our data, Minegishi et al. showed that *Cyp26b1* is selectively expressed in the upper zones of growth plate to keep endogenous retinoids at very low levels and that its conditional ablation in cartilage caused a significant drop in chondrocyte proliferation and derangement of growth plate zonal organization.⁽³⁰⁾ We reported a similar drop in chondrocyte proliferation and abnormal growth plate structure and functioning in mice in which we conditionally deleted *RAR α* and *RAR γ* in cartilage.⁽³²⁾ The drop in *Cyp26b1* expression following LDE225 treatment combined with broader and higher *Raldh3* expression observed here is thus likely to have caused a major overall increase in -and broader distribution of-active retinoids, deranging growth plate homeostasis, accelerating maturation and hypertrophy and ultimately causing precocious closure (see our model in Fig. 8). Relevantly also, it was recently found that hedgehog signaling is normally needed for embryonic expression of *Cyp26* family members⁽⁵⁹⁾, suggesting that the same may occur in growth plate and that LDE225 could have acted directly to down-regulate *Cyp26b1* expression. Together, the studies support the overall notion that in addition to hedgehog signaling and other mechanisms⁽²⁾, the retinoid pathway is intimately interwoven into growth plate biology, contributes to fine-tune its activities and roles, and can have major phenotypic effects once genetically or pharmacologically modulated. We should note that mutual interactions between the retinoid and hedgehog pathways have been demonstrated in a number of developmental processes and systems, with interactions being negative or positive depending on biological context and developmental stage.^(59–62) Within the growth plate, the specific retinoid action is dictated by lack or presence of ligands as indicated above, thus allowing this pathway to exert distinct zone-specific effects in relation to other pathways -including the hedgehog

pathway- and contribute to the regulation of overall growth plate physiology.⁽⁶³⁾ Our in vitro data reiterate the potency of the hedgehog and retinoid pathways in dictating and coordinating both chondrogenic cell differentiation and chondrocyte terminal maturation. The complex phenotypes displayed by the cells during drug co-treatment -such as low *Mef2c* expression and high *Ihh* expression compared to control- indicate that chondrogenic cells can modulate those pathways in independent manners to adjust to given circumstances. It remains to be determined whether similar subtle changes in regulatory interactions between these pathways occur in vivo and could adjust the responses of growth plate and the skeleton in general to diverse normal and pathological conditions.

Many complex diseases including cancer and musculoskeletal disorders involve changes in a spectrum of cellular and molecular regulatory mechanisms.⁽⁶⁴⁾ This is of course not surprising given the multitude of cell-cell and tissue-tissue interactions and local and systemic interplays that govern physiology and the fact that disease culprits can affect different tissues and organs in distinct manners. As elaborated in a recent incisive review, many such diseases are currently treated with single-target monotherapy, and drug development programs are often directed on identifying mono-therapeutic regimens.⁽⁵¹⁾ To increase effectiveness, a number of combination drug therapies have been conceived, tested and implemented in several areas of clinical medicine, with measures of success particularly in cancer.⁽⁵¹⁾ By definition, combination therapies concurrently target distinct pathogenic pathways, could be efficacious at drug doses lower than respective monotherapies, could even be adjusted to meet patient-to-patient variability, or could address unwanted side effects. The latter describes the drug combination in our present proof-of-principle study. For such co-treatment to work, it would of course be necessary to establish that the RAR γ antagonist does not in itself affect the pharmacokinetics of LDE225 and its efficacy on targeted cancer cells. Notably, previous studies described beneficial therapeutic effects of other retinoid antagonists against malignant cells including breast and prostate cancer cells.^(65–67) Thus, it is possible that a combined LDE225/ RAR γ antagonist regimen may be more efficacious against certain cancers. Future research is needed to address these and related questions, but our data do identify intriguing therapeutic possibilities for the future.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgements

This study was supported by NIH/NIAMS grant RO1AR056837 (to MI and MP).

Each author acknowledges to be fully compliant with institutional mandates on responsible conduct in research, animal welfare, human subjects and laboratory safety compliance pertaining to the data and procedures described in the manuscript.

Data accessibility statement

All data forming the basis of the present study are presented in the manuscript.

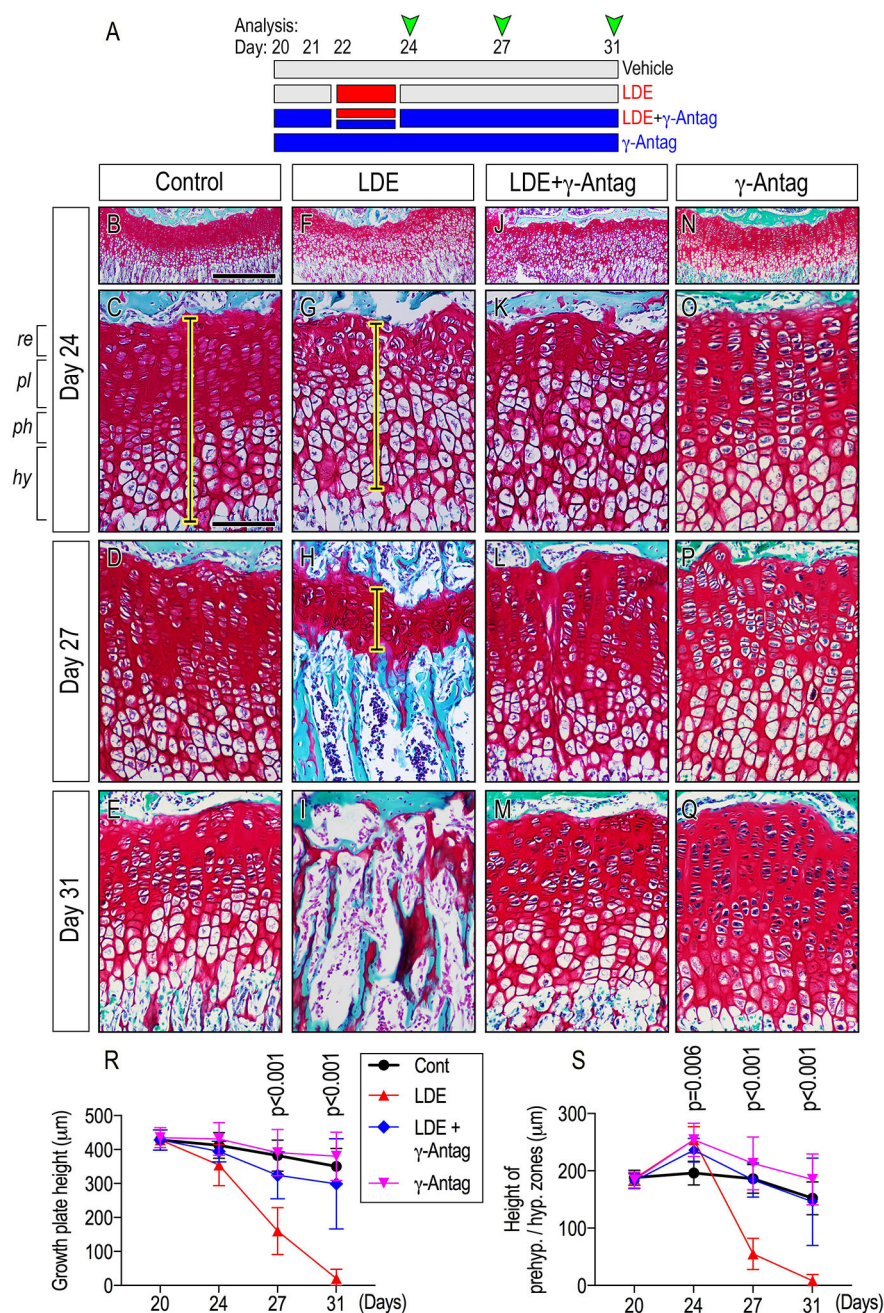
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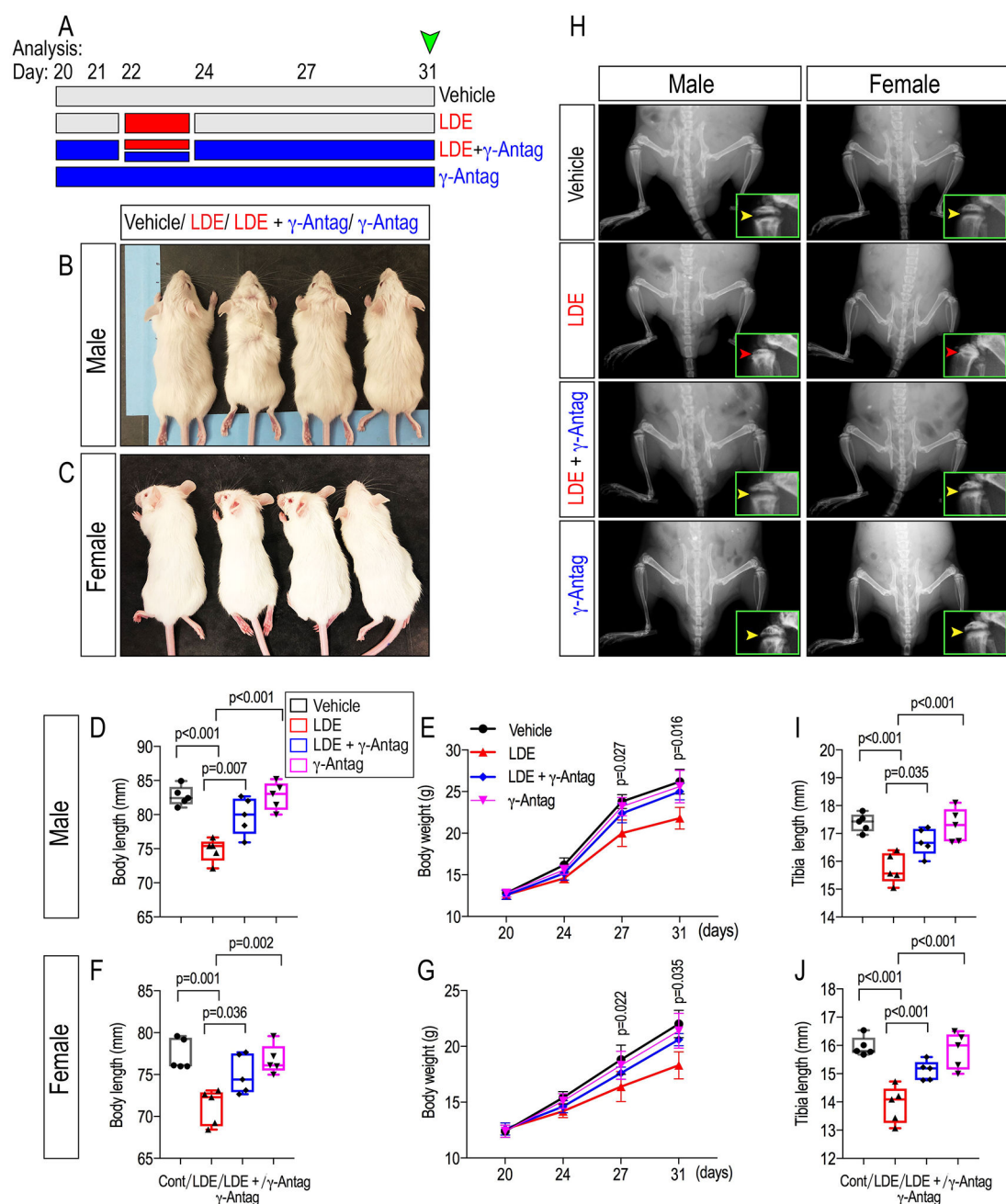
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**Fig. 1.**

Tibial growth plate is deranged by LDE225 treatment but is maintained by drug co-treatment. (A) Schematic of experimental plan starting with P20 mice, with control mice receiving vehicle and treated mice receiving LDE225, RAR γ antagonist (γ -Antag) or LDE225/RAR γ -antagonist combination. Green arrowheads point to sequential days of harvest. (B-E) Representative images of control safranin-O stained sections of proximal tibial growth plates harvested at indicated mouse ages shown at low mag (B) and higher mag (C-E). Approximate location of resting (*re*), proliferative (*pr*), pre-hypertrophic (*ph*) and hypertrophic (*hy*) zones is indicated by brackets. Vertical yellow line in (C) indicates

overall control growth plate height. (*F-I*) Images of stained tibial growth plate sections from LDE225-treated mice. Vertical lines in (G) and (H) depict the progressive and rapid growth plate shortening that is completed by P31 (I). (*J-M*) Images of stained tibial growth plate sections from LDE225/ γ -Antag co-treated mice. (*N-Q*) Images of stained tibial growth plate sections from γ -Antag-treated mice. (*R-S*) Linear plots depicting average heights of growth plates (R) and pre-hypertrophic plus hypertrophic zones (S) in control versus treated mice at successive days until P31. Days 24, 27 and 31: n=5 (cont), n=5 (LDE225 or γ -Antag) and n=5 (LDE225/ γ -Antag) mice. One-way ANOVA followed by Tukey's multiple comparison test. *p*, LDE2225 vs all other groups. Scale bar in (B) for (B, F, J and N) = 550 μ m; scale bar in (C) for all other panels, 150 μ m.

**Fig. 2.**

Body length and weight are preserved in co-treated mice. (A) Schematic of experimental plan and harvest time at P31. (B-G) General appearance of (B) male and (C) female control, LDE225-treated, γ -Antag-treated and LDE225/ γ -Antag co-treated mice and their respective body length (D and F) and body weight (E and G) over experimental time until P31. (H) X-ray images of hind limbs in P31 male and female control, treated and co-treated mice and the respective measured lengths of their tibias (I-J). Yellow arrowheads inside the green box insets point to the growth plate. n=5 male and n=5 female mice per group. One-way ANOVA followed by Tukey's multiple comparison test. Horizontal brackets indicate *p*

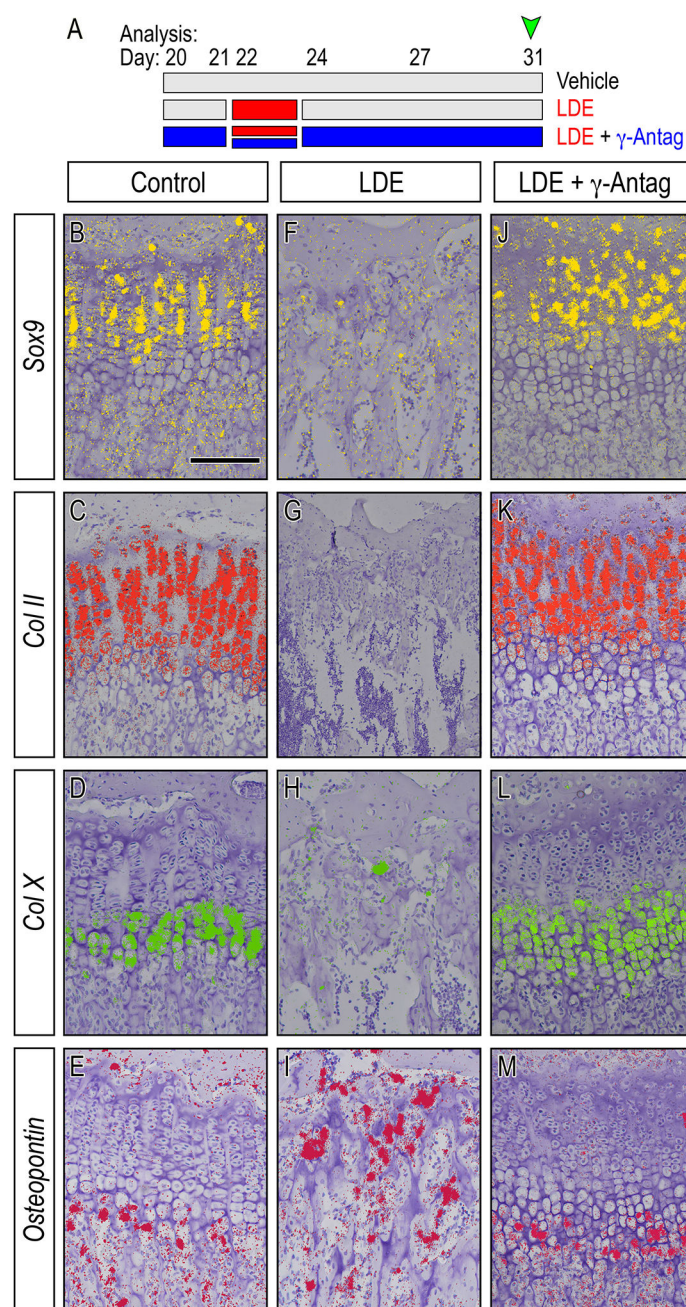
values between given groups. In E and G, p values are for LDE225 group versus all other groups.

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**Fig. 3.**

Expression patterns of characteristic genes demarcate growth plate zones and phenotypic changes. (A) Schematic of experimental plan. (B-E) Representative in situ hybridization images of P31 tibial growth plate sections from control vehicle-receiving mice depicting expression of: (B) *Sox9*; (C) collagen II (*Col II*); (D) collagen X (*Col X*); and (E) *osteopontin*. In situ hybridization was carried out using radiolabeled riboprobes, and hybridization signal was pseudo-colored using Image-J. (F-I) In situ hybridization images of P31 tibia sections from P31 LDE225-treated mice showing: the near absence of *Sox9*, *Col II* and *Col X* expression concomitant with growth plate involution (F-H); and widespread

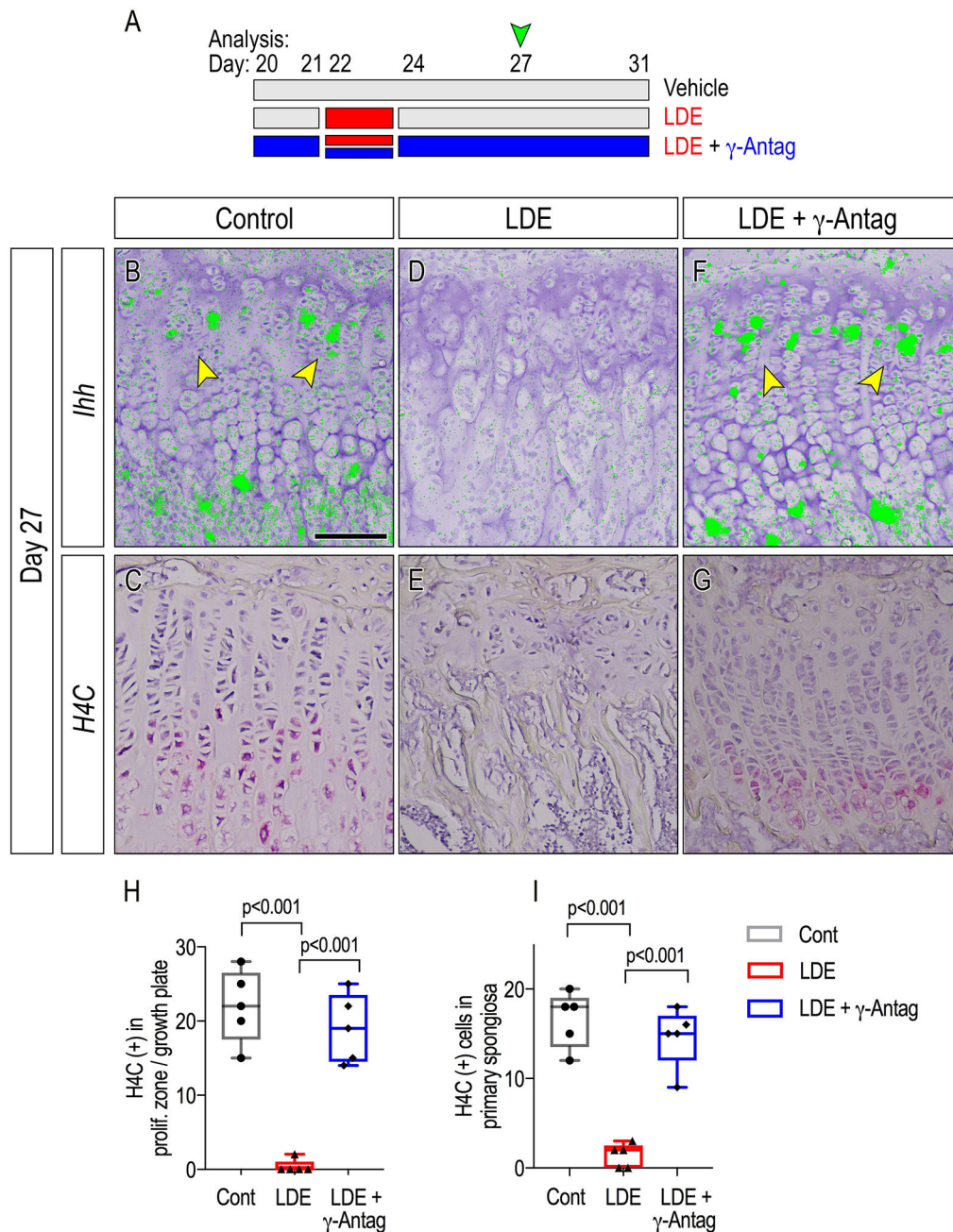
expression of *osteopontin* within the osseous tissue replacing the growth plate (I). (*J-M*) In situ hybridization images of P31 tibial growth plate sections from P31 LDE225/ γ -Antag co-treated mice showing gene expression pattern preservation. Scale bar = 150 μ m.

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**Fig. 4.**

Proliferative zone of growth plate in relation to *Ihh* expression and drug treatment. (A) Schematic of experimental plan. (B-C) Representative in situ hybridization images of expression patterns of the mitotic marker gene *H4C* and of *Ihh* in P27 control tibial growth plate. Arrowheads point to *H4C*-positive chondrocytes in proliferative zone. There are also several *H4C*-expressing cells in primary spongiosa. (D-G) Expression patterns of the two genes in tibial growth plates of P27 LDE-225-treated mice (D-E) and LDE225/ γ -Antag co-treated mice (F-G). (H-I) Quantification of *H4C*-positive cells in growth plate (H) and primary spongiosa (I) in P27 control and treated and co-treated mice. n=5 mice per

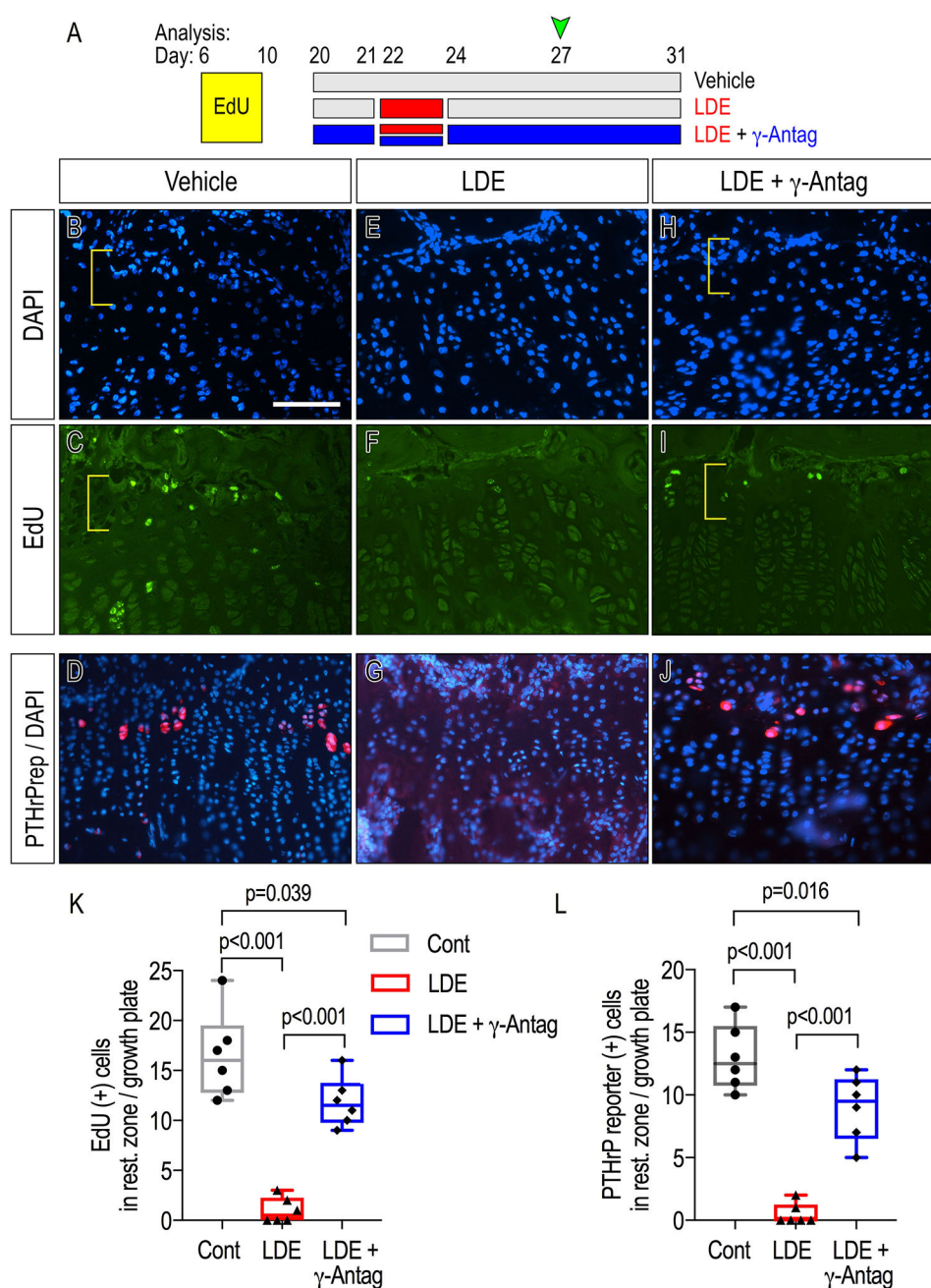
group. One-way ANOVA followed by Tukey's multiple comparison test. Horizontal brackets indicate p values between compared groups. Scale bar = 150 μm .

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**Fig. 5.**

Slow cycling cells in the reserve zone and relationship to *Pthrp* expression. (A) Schematic of experimental plan. (B-D) EdU-labeled slow cycling cells localized in the reserve zone (brackets in B-C) and expression of *Pthrp*-reporter in similar zone counterstained with DAPI (D). (E-J) Patterns of EdU-labeled cell distribution and *Pthrp*-reporter expression in P27 LDE-225-treated mice (E-G) and LDE225/ γ -Antag co-treated mice (H-J). Brackets in (H and I) point to persistence of EdU-labeled cells in co-treated growth plates. (K-L) Quantification of EdU-positive cells (K) and *Pthrp*-reporter expressing cells (L) in P27 control and treated and co-treated mice. n=3 EDU-labeled mice and n=3 *Pthrp*-reporter

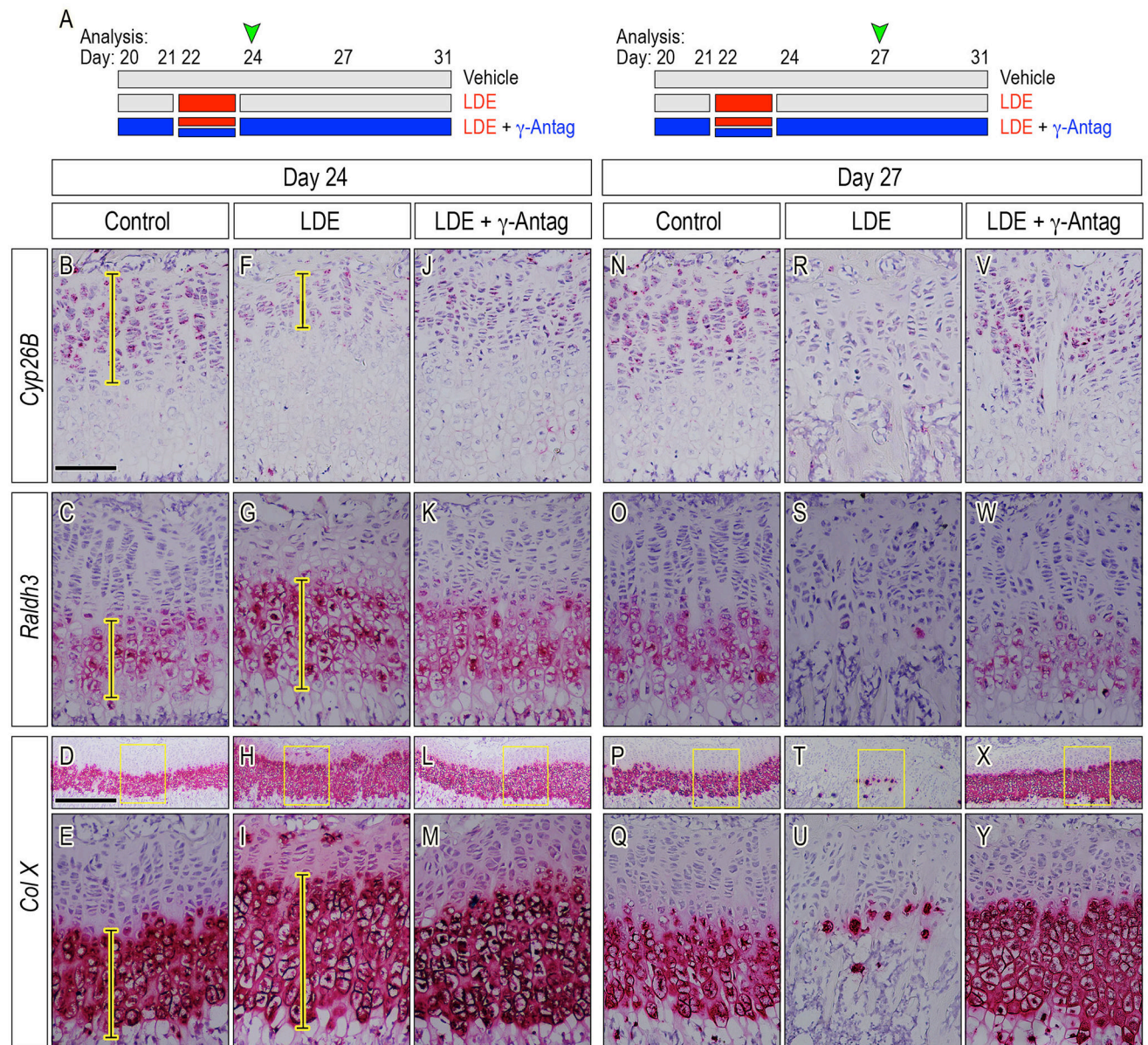
mice per group. Both left and right sides of tibial growth plates were analyzed. One-way ANOVA followed by Tukey's multiple comparison test. Horizontal brackets indicate p values between compared groups. Scale bar = 250 μm .

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**Fig. 6.**

Gene expression patterns of retinoid metabolic enzymes shift during LDE225 treatment. (A) Schematic of experimental plans. (B-E and N-Q) Representative RNAscope in situ hybridization images of *Cyp26b1*, *Raldh3* and *Col X* in control P24 (B-E) and P27 (N-Q) tibial growth plates. *Cyp26b1* expression characterizes the upper growth plate zones, while *Raldh3* expression is restricted to the hypertrophic zone and overlaps *Col X* expression in the hypertrophic zone. Vertical yellow lines depict the approximate heights of upper zones and hypertrophic zone. (F-I and R-U) In situ images of expression patterns of the 3 genes in LDE225-treated P24 (F-I) and P27 (R-U) mice. At P24, *Cyp26b1* expression has become narrower (F, vertical line), while *Raldh3* expression is clearly broader and expanded upward along with a thicker *Col X*-expressing hypertrophic zone (G-I, vertical lines). By P27 (R-U), expression of the 3 genes is barely detectable in growth plate remnants. (J-M and V-Y)

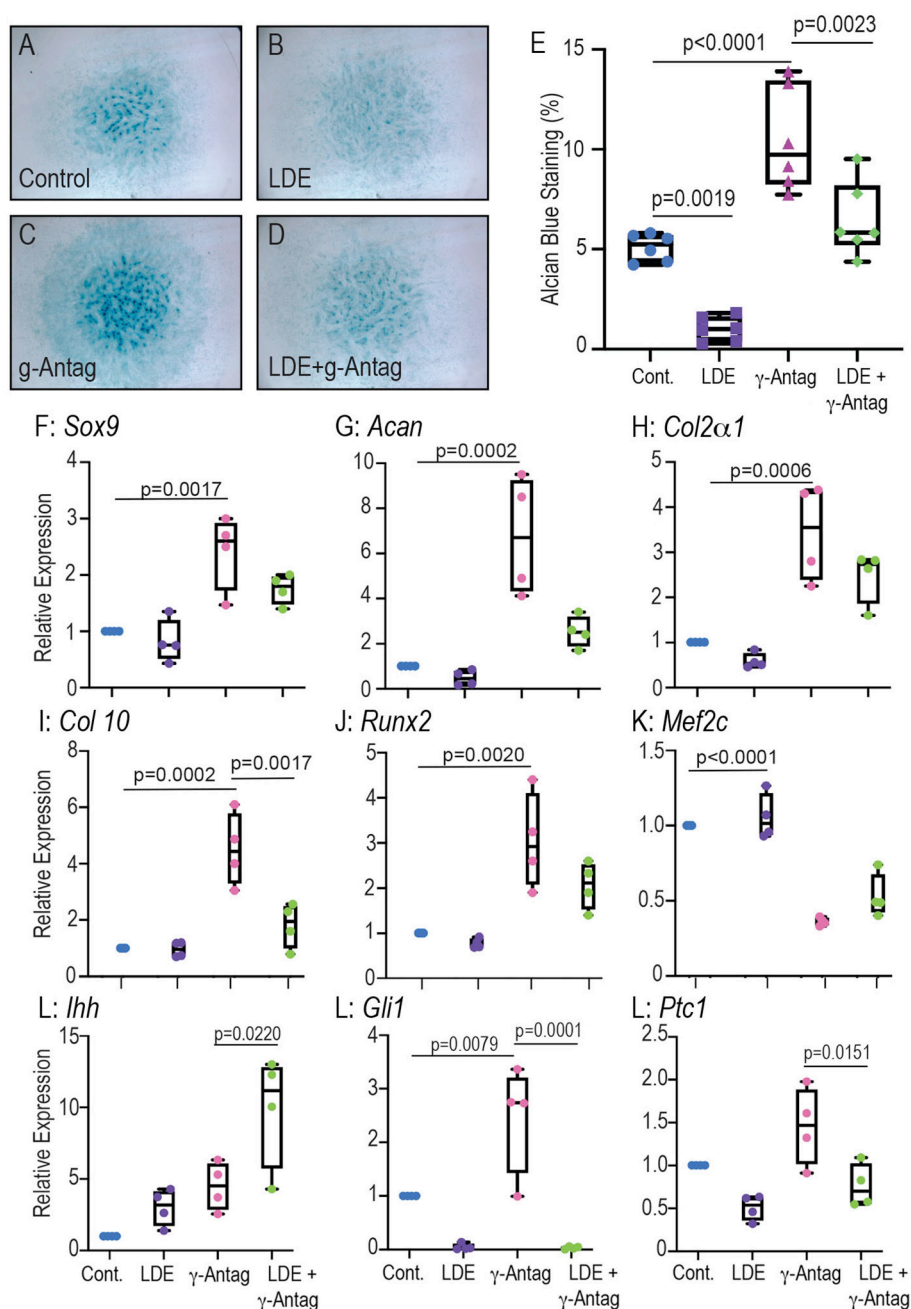
Images showing the expression patterns of the 3 genes in P24 (J-M) and P27 (V-Y) growth plates from LDE225/ γ -Antag co-treated mice that are similar to those in controls. Each low mag area indicated by a yellow square box in D, H, L, P, T and X is shown at higher magnification in image immediately below. Scale bar in (B) for all high mag images = 150 μ m; scale bar in (D) for (D, H, L, P, T and X) = 550 μ m.

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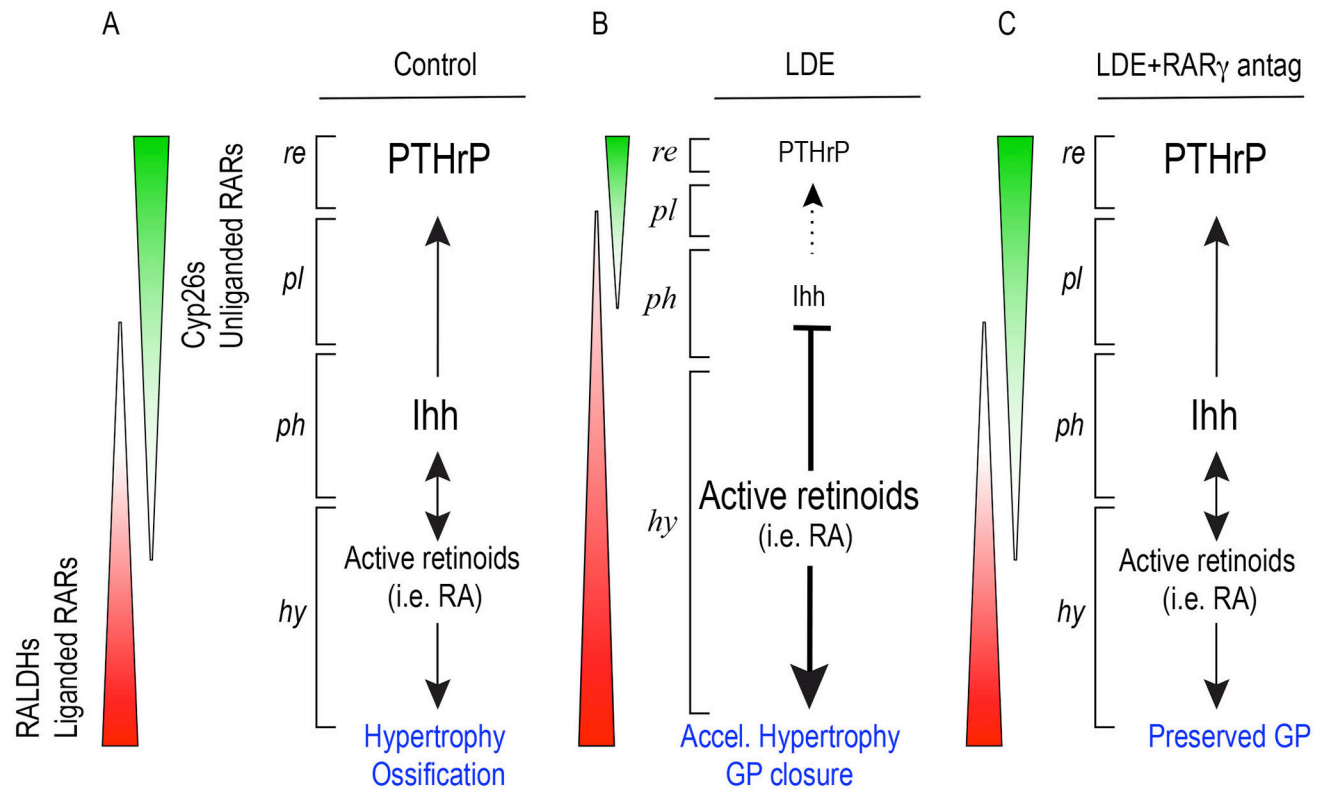
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**Fig. 7.**

Chondrogenesis in vitro is directly modulated by drug treatment and co-treatment. (A-D) Representative images of alcian blue-stained limb bud cell micromass cultures on day 6 that had received continuous treatment with vehicle (A), LDE (B), γ -Antag (C) or LDE225/ γ -Antag (D). (E) Quantification of alcian blue staining in multiple control and treated micromass cultures. (F-N) Total RNA was isolated from day 6 cultures and assessed by quantitative PCR for: (F-H) *Sox9*, *Col II* (*Col2 α 1*) and *aggrecan* (*Acan*); (I-K) *ColX*, *Runx2* and *Mef2c*; and (L-N) *Ihh*, *Ptc1* and *Gli1*. Cell culture experiments were

repeated independently at least three times. One-way ANOVA followed by Tukey's multiple comparison test. Horizontal brackets indicate p values between sample groups.

**Fig. 8.**

Working model of growth plate mechanisms operating in control and drug-treated mice.

(A) In control conditions, the Ihh-PTHrP axis sets the rates of chondrocyte proliferation and maturation within upper growth plate zones that express *Cyp26s*, contain minimal endogenous retinoids in a downward gradient (green triangle) and are regulated by unliganded RARs exerting repressor function. Transition to hypertrophy would activate *Raldh* expression and production of biologically active retinoids and mainly *all-trans*-retinoic acid (RA), creating an upward gradient (red triangle) and enabling liganded RARs to exert transcriptional activity and promote terminal chondrocyte maturation and transition to endochondral bone. Retinoid action and Ihh signaling would normally interact, balance and restrain each other, maintaining a physiologic dynamic boundary between upper zones and hypertrophic zone. (B) By interfering with hedgehog signaling, LDE225 would destabilize growth plate homeostasis by directly or indirectly favoring *Raldh* expression, expanding the levels, distribution and action of RA and other active retinoids. In turn, this would dampen the Ihh-PTHrP axis, narrow the upper zones, accelerate chondrocyte maturation and hypertrophy and ultimately cause growth plate closure. (C) Co-treatment with RAR γ antagonist would largely counter the acceleration of hypertrophy and ossification, restore more normal *Ihh* and *Pthrp* expression and reciprocal restraining interactions amongst the zones, and allow the growth plate to maintain organization and function. *Rz*, *pl*, *ph* and *hy*: resting, proliferative, pre-hypertrophic and hypertrophic zones, respectively; GP, growth plate.